Angle-Based Formation Tracking of Underactuated Planar Agents

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Outline

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Motivation

Formation control of multi-agent systems [1]

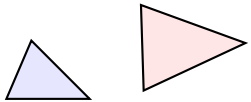
- UAV swarms, satellite formations, multi-robot platforms, marine surface vessels

How is the desired shape described?

- Displacement [8], distance [10], ratio-of-distances [2]
- Bearing [15, 13]
- Angle [9, 4]

Why angle-based?

- Angles are invariant to **translation, rotation and scaling**
- Only **local** measurements needed, no common reference frame



same angles, different pose & size

Related Work and Research Gap

- Angle rigidity: Eren [7], Jing et al. [9] (local stability) → **triangular angle rigidity** [3] (global stability)

Work	Agent model	Remarks
Chen et al. [5]	Single integrator	Distributed angle-based law
Chen et al. [6]	Double integrator	PD-type law, needs relative velocity
Peng et al. [12]	Double integrator	Matched disturbances, internal model
Lu et al. [11]	Underactuated vessel	Star graph only (leader at center)
This work	**Underactuated planar**	**Angle-based, disturbances**

Gap

Angle-based formation control for **underactuated** agents has not been studied.

Contributions

1. First angle-based formation tracking framework for **underactuated planar agents** subject to **disturbances**, using a variable structure control approach
2. **Control law 1** (with relative velocity): exponential formation tracking with internal stability guarantees
3. **Control law 2** (without relative velocity): first angle-based law for relative-degree-two agents with additive disturbances that needs only relative positions
4. Validation by simulations on a group of underactuated surface vessels (USVs)

1 Introduction

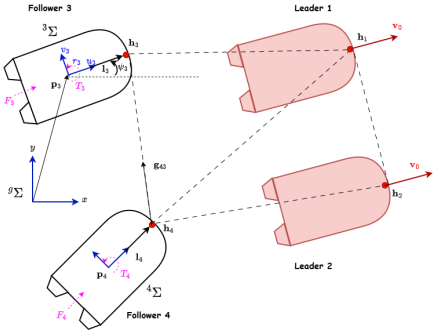
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Underactuated Planar Agent Model



Two leaders and two followers

Kinematics

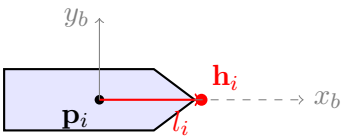
$$\dot{\mathbf{p}}_i = \mathcal{R}(\psi_i) \begin{bmatrix} u_i \\ v_i \end{bmatrix}, \qquad \dot{\psi}_i = r_i$$

Dynamics

$$\begin{cases} \dot{u}_i = g_{ui} + \delta_{ui}(t) + b_{ui} F_i \\ \dot{v}_i = g_{vi} + \delta_{vi}(t) + b_{vi} T_i \\ \dot{r}_i = g_{ri} + \delta_{ri}(t) + b_{ri} T_i \end{cases}$$

- 3 DOF (surge, sway, yaw), only **2 inputs**: surge force F_i, yaw moment T_i
- Sway and yaw both driven by $T_i \Rightarrow$ underactuated
- $\delta_{(\cdot)i}$: bounded disturbances; example: USV model

The Hand Point



$\mathbf{h}_i = \mathbf{p}_i + \mathbf{l}_i$, $\mathbf{l}_i$ along body x-axis

Velocity and acceleration of $\mathbf{h}_i$ in the body frame [14]:

$$ {}^{i}\mathbf{v}_{hi} = \begin{bmatrix} u_i \\ v_i + r_i l_i \end{bmatrix} $$

$$ {}^{i}\mathbf{a}_{hi} = \begin{bmatrix} g_{ui} + \delta_{ui} - r_i^2 l_i + b_{ui} F_i \\ g_{vi} + l_i g_{ri} + \delta_{vi} + l_i \delta_{ri} + (b_{vi} + b_{ri} l_i) T_i \end{bmatrix} $$

Why the hand point?

Each input acts on one component of ${}^{i}\mathbf{a}_{hi} \Rightarrow$ the hand point is **fully actuated**:

$$ \ddot{\mathbf{h}}_i = \boldsymbol{\nu}_i + \boldsymbol{\zeta}_i \quad \text{(double integrator + disturbance)} $$

Triangular Angle Rigidity and Problem Statement

- Angle α_{kij}: rotate $\mathbf{g}_{ik}$ counterclockwise to $\mathbf{g}_{ij}$, $\mathbf{g}_{ij} = \frac{\mathbf{h}_j - \mathbf{h}_i}{\|\mathbf{h}_j - \mathbf{h}_i\|}$
- Each triangle $\triangle ijk$ gives a linear constraint [3]:

$$\mathbf{A}_i\mathbf{h}_i + \mathbf{A}_j\mathbf{h}_j + \mathbf{A}_k\mathbf{h}_k = \mathbf{0}$$

- Stacked: $\mathbf{R}_{\mathcal{A}}(\alpha)\mathbf{h} = \mathbf{0}$; triangularly angle rigid $\Leftrightarrow \operatorname{rank}\mathbf{R}_{\mathcal{A}} = 2n-4$

- Leader–follower graph, $n_l \geq 2$ leaders
- Leaders start at desired positions, move with constant $\mathbf{v}_0$

Lemma ([5])

With $\mathbf{L} = \mathbf{R}_{\mathcal{A}}^{\top}\mathbf{R}_{\mathcal{A}}$, $\mathbf{L}_{ff} \succ 0$ *and*

$$\mathbf{h}^{F*} = -\mathbf{L}_{ff}^{-1}\mathbf{L}_{fl}\,\mathbf{h}^{L}, \qquad \dot{\mathbf{h}}^{F*} = \mathbf{1}\otimes\mathbf{v}_0$$

Problem

Design F_i, T_i *for each follower such that* $\mathbf{h}(t) \to \mathbf{h}^*(t)$ *as* $t \to \infty$.

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Control Law 1: With Relative Velocity

Sliding variable (velocity + PI action on the angle error):

$$ {}^{i}\mathbf{s}_i = \mathcal{R}_i^{\top}\left[\dot{\mathbf{h}}_i + k_P\,\mathcal{F}_i(\alpha^*, \mathbf{h}) + k_I \int_0^t \mathcal{F}_i(\alpha^*, \mathbf{h}(\tau))\,d\tau\right] $$

$\mathcal{F}_i$: i-th block of $\mathbf{R}_{\mathcal{A}}^{\top}(\alpha^*)\mathbf{R}_{\mathcal{A}}(\alpha^*)\mathbf{h}$, computed from neighbours' relative positions

Control law:

$$ \begin{bmatrix} b_{ui}F_i \\ (b_{vi} + b_{ri}l_i)T_i \end{bmatrix} = -k\,{}^{i}\mathbf{s}_i - \gamma\,\mathrm{sgn}({}^{i}\mathbf{s}_i) - \begin{bmatrix} g_{ui} - r_i^2 l_i \\ g_{vi} + l_i g_{ri} \end{bmatrix} - \mathcal{R}_i^{\top}\left(k_P\mathcal{F}_i(\alpha^*, \mathbf{v}_h) + k_I\mathcal{F}_i(\alpha^*, \mathbf{h})\right) $$

Theorem

If $\gamma > \left\|[\delta_{ui},\ \delta_{vi} + l_i\delta_{ri}]\right\|_{\infty}$, *then* $\mathbf{h}(t) \to \mathbf{h}^*(t)$ *exponentially.*

Proof idea: $V = \frac{1}{2}\|{}^{i}\mathbf{s}_i\|^2 \Rightarrow \dot{V} \le -k\|{}^{i}\mathbf{s}_i\|^2$, so $\mathbf{s}_i \to \mathbf{0}$ exponentially; the formation error then obeys a Hurwitz linear system driven by a vanishing input.

Needs: relative positions **and relative velocities**

Control Law 2: Without Relative Velocity

Motivation: relative velocity is usually obtained by differentiating noisy position measurements.

Feedback linearization: $\ddot{\mathbf{h}}_i = \boldsymbol{\nu}_i + \boldsymbol{\zeta}_i, \quad \boldsymbol{\zeta}_i = \mathcal{R}_i[\delta_{ui},\ \delta_{vi} + l_i\delta_{ri}]^\top$

Virtual control law, with $\mathbf{s}_i = -\mathcal{F}_i(\alpha^*, \mathbf{h})$:

$$\boldsymbol{\nu}_i = \mathbf{s}_i + \dot{\boldsymbol{\eta}}_i + \gamma\,\mathrm{sgn}(\boldsymbol{\eta}_i - \mathbf{v}_{hi})$$
$$\dot{\boldsymbol{\Upsilon}}_i = -k_1(\boldsymbol{\Upsilon}_i - \boldsymbol{\eta}_i)$$
$$\dot{\boldsymbol{\eta}}_i = k_2\mathbf{s}_i + k_1k_3(\boldsymbol{\Upsilon}_i - \boldsymbol{\eta}_i)$$

- $\boldsymbol{\eta}_i$: velocity reference generated from position errors
- Sign term: $\mathbf{v}_{hi}$ tracks $\boldsymbol{\eta}_i$ despite disturbances
- Filter $\boldsymbol{\Upsilon}_i$: adds damping

Theorem

If $\boldsymbol{\zeta}(t)$ is uniformly continuous and $\gamma > \sqrt{2}\left\|[\delta_{ui},\ \delta_{vi} + l_i\delta_{ri}]\right\|_\infty$, then $\mathbf{h}(t) \to \mathbf{h}^*(t)$ as $t \to \infty$.

Proof idea: Lyapunov function + Barbalat's lemma (twice).

Needs: relative positions + **own** velocity only

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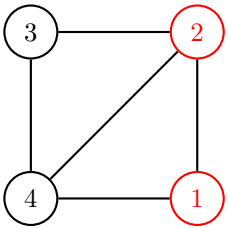
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Simulation Setup: Four USVs



Leaders 1, 2 (red); followers 3, 4

$q = 2$ triangles

$$\alpha^*_{421} = \frac{\pi}{4},\ \alpha^*_{214} = \frac{\pi}{2}$$
$$\alpha^*_{324} = \frac{\pi}{4},\ \alpha^*_{243} = \frac{\pi}{4}$$

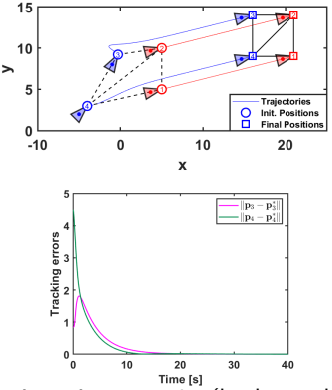
- Leaders: $\mathbf{h}_1(0) = [5, 5]^\top$, $\mathbf{h}_2(0) = [5, 10]^\top$, $\mathbf{v}_0 = [0.4, 0.1]^\top$
- Homogeneous USVs, hand point offset $l_i = 1.4\,\text{m}$
- Disturbances:
  - $w_{ui} = 50 + 30\sin(2t) + 30\sin(10t)$ N
  - $w_{ri} = 100 + 60\sin(2t) + 40\sin(10t)$ Nm

Control law	Gains
Law 1	$k = 3,\ k_P = 1.5,\ k_I = 0.15,\ \gamma = 2$
Law 2	$k_1 = 0.5,\ k_2 = k_3 = 5,\ \gamma = 5$

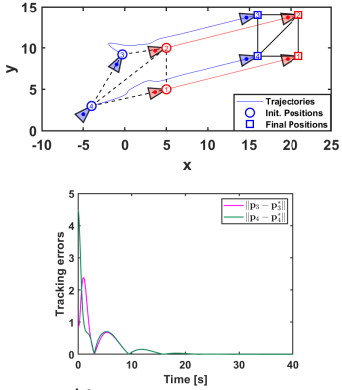
Vessel parameters: see backup slide

Simulation Results

Law 1 (with relative velocity)



Law 2 (without relative velocity)



Top: trajectories over 40 s (leaders red, followers blue). Bottom: tracking errors.

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Conclusion and Future Work

- Two distributed **angle-based** formation tracking laws for **underactuated** planar agents with disturbances
- Formation defined on hand points, triangularly angle rigid; leaders move with constant velocity

	Law 1	Law 2
Relative velocity	Required	Not required
Convergence	Exponential	Asymptotic
Analysis	Sliding variable + Hurwitz system	Lyapunov + Barbalat

Future work

- Maneuvering leaders (translation, rotation, scaling)
- Adaptive gains when disturbance bounds are unknown

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Thank you and Questions



$$\mathcal{M}_i \begin{bmatrix} \dot{u}_i \\ \dot{v}_i \\ \dot{r}_i \end{bmatrix} + \begin{bmatrix} -m_{22,i} v_i r_i - m_{23,i} r_i^2 \\ m_{11,i} u_i r_i \\ (m_{22,i} - m_{11,i}) u_i v_i + m_{23,i} u_i r_i \end{bmatrix} + \mathcal{D}_i \begin{bmatrix} u_i \\ v_i \\ r_i \end{bmatrix} = \begin{bmatrix} F_i \\ 0 \\ T_i \end{bmatrix} + \begin{bmatrix} w_{ui} \\ w_{vi} \\ w_{ri} \end{bmatrix}$$

$$\mathcal{M}_i = \begin{bmatrix} m_{11,i} & 0 & 0 \\ 0 & m_{22,i} & m_{23,i} \\ 0 & m_{23,i} & m_{33,i} \end{bmatrix}, \qquad \mathcal{D}_i = \begin{bmatrix} d_{11,i} & 0 & 0 \\ 0 & d_{22,i} & d_{23,i} \\ 0 & d_{23,i} & d_{33,i} \end{bmatrix}$$

Multiplying by $\mathcal{M}_i^{-1}$ gives the general model.

$m_{11,i}$	220 kg	$d_{11,i}$	75 kg/s
$m_{22,i}$	295 kg	$d_{22,i}$	105 kg/s
$m_{33,i}$	745 kg m^2	$d_{33,i}$	50 kg m^2/s
$m_{23,i}$	3 kg m	$d_{23,i}$	-25 kg m/s

Backup: Internal Stability (Law 1)

- Theorem 1 gives bounded $\dot{\mathbf{h}}$; since $\mathcal{R}_i^\top \dot{\mathbf{h}}_i = [u_i,\ v_i + r_i l_i]^\top$, u_i is bounded
- Boundedness of $v_i \Leftrightarrow$ boundedness of $r_i \Rightarrow$ suffices to bound the sway velocity

Remark

For the USV model, v_i remains bounded (following [14]) if

$$\|\mathbf{v}_0\|_\infty < \frac{d_{22,i} l_i - d_{23,i}}{m_{11,i}} \quad \textit{and} \quad l_i > \max\left\{\frac{d_{23,i}}{d_{22,i}},\ \frac{m_{23,i}}{m_{22,i}}\right\}.$$

Backup: Proof Sketch of Theorem 2

Lyapunov function, with $\boldsymbol{\delta}^F = \mathbf{h}^F - \mathbf{h}^{F*}$:

$$V = \tfrac{1}{2}(\boldsymbol{\delta}^F)^\top \mathbf{L}_{ff}\boldsymbol{\delta}^F + \tfrac{1}{2}\|\boldsymbol{\eta}^F - \mathbf{v}_h^F\|^2 + \tfrac{1}{2k_2}\|\boldsymbol{\eta}^F - \mathbf{1}\otimes\mathbf{v}_0\|^2 + \tfrac{k_3}{2k_2}\|\boldsymbol{\Upsilon}^F - \mathbf{1}\otimes\mathbf{v}_0\|^2$$

- $\dot{V} \le -(\gamma - \|\boldsymbol{\zeta}\|_\infty)\|\boldsymbol{\eta}^F - \mathbf{v}_h^F\|_1 - \frac{k_1 k_3}{k_2}\|\boldsymbol{\Upsilon}^F - \boldsymbol{\eta}^F\|^2 \le -\frac{k_1 k_3}{k_2}\|\boldsymbol{\Upsilon}^F - \boldsymbol{\eta}^F\|^2$
- $\sqrt{2}$ factor: $\|\mathcal{R}\mathbf{x}\|_\infty \le \sqrt{2}\|\mathbf{x}\|_\infty$ for any rotation $\mathcal{R}$
- Barbalat's lemma $\Rightarrow \boldsymbol{\Upsilon}^F - \boldsymbol{\eta}^F \to \mathbf{0}$, applied again $\Rightarrow \dot{\boldsymbol{\eta}}^F \to \mathbf{0}$
- Hence $\mathbf{s}^F \to \mathbf{0}$, i.e. $\mathbf{h}(t) \to \mathbf{h}^*(t)$